

Topological insulators

From occupied bands to a protected edge

Two insulators can have the same symmetries and still be impossible to connect without closing the bulk gap. For noninteracting electrons, the obstruction resides in the occupied Bloch states. This note contrasts a Chern insulator with a time-reversal-invariant topological insulator, then tests edge protection with a two-component Hamiltonian.

1 The geometry of an occupied band

Let $|u_n(\mathbf{k})\rangle$ be the cell-periodic part of an isolated occupied Bloch band in two dimensions. Define its Berry connection and curvature by

$$\mathcal{A}_{n,i} = i\langle u_n | \partial_{k_i} u_n \rangle, \quad \Omega_n = \partial_{k_x} \mathcal{A}_{n,y} - \partial_{k_y} \mathcal{A}_{n,x}. \quad (1)$$

Changing the phase to $e^{i\chi_n(\mathbf{k})}|u_n\rangle$ changes $\mathcal{A}_n$ by $-\nabla_{\mathbf{k}}\chi_n$, but leaves Ω_n unchanged. The occupied-band Chern number is

$$C = \frac{1}{2\pi} \sum_{n \in \text{occ}} \int_{\text{BZ}} d^2k \Omega_n(\mathbf{k}) \in \mathbb{Z}. \quad (2)$$

Here BZ is the Brillouin-zone torus. For degenerate occupied bands, one uses the trace of the occupied-subspace non-Abelian curvature instead of assigning separate band curvatures. Integer quantization follows from how smooth phase choices are patched between regions of the torus.

The integer cannot change continuously while the occupied and empty subspaces remain separated by a gap. With a consistent orientation convention, the Hall conductivity is $\sigma_{xy} = Ce^2/h$, where $e > 0$ is the elementary charge. A nonzero C requires broken time reversal for ordinary charge-conserving electronic bands.

2 Zero Chern number does not mean trivial

For spinful electrons, the time-reversal operator Θ satisfies $\Theta^2 = -1$. The total occupied curvature is odd under $\mathbf{k} \mapsto -\mathbf{k}$, so its integral vanishes. Nevertheless, the occupied states can retain a $\mathbb{Z}_2$ obstruction [1, 2].

An instructive special case conserves a spin component: two time-reversed blocks have $C_{\uparrow} = -C_{\downarrow}$, and the two-dimensional invariant is

$$\nu = C_{\uparrow} \pmod{2} \in \{0, 1\}. \quad (3)$$

Turning on spin mixing destroys the separate spin-block Chern numbers, but preserves ν as long as time reversal and the bulk gap survive. Thus a quantized spin Hall conductivity is not the general definition of this phase. The robust statement is the parity of boundary Kramers pairs.

3 A minimal edge calculation

Near one edge crossing, use a basis of two counterpropagating Kramers partners. Let k now denote the one-dimensional edge momentum measured from the crossing, and let the Pauli matrices act on this pair:

$$H_{\text{edge}}(k) = \hbar v k \sigma_z, \quad \Theta = i\sigma_y \mathcal{K}. \quad (4)$$

Here $v > 0$ is the edge velocity and $\mathcal{K}$ denotes complex conjugation. Since $\Theta\sigma_i\Theta^{-1} = -\sigma_i$, the Hamiltonian obeys $\Theta H(k)\Theta^{-1} = H(-k)$.

A constant term $m_x\sigma_x + m_y\sigma_y$ would open a gap:

$$E_{\pm}(k) = \pm \sqrt{(\hbar v k)^2 + m_x^2 + m_y^2}. \quad (5)$$

But both mass terms are odd under time reversal, so they are forbidden if that symmetry is preserved. A scalar term shifts the crossing energy without opening a gap. This is the simplest local explanation of protection for one isolated helical pair.

Two pairs can instead gap without breaking time reversal. For example, introduce Pauli matrices τ_i labeling the pairs, take $\Theta = i\sigma_y \mathcal{K}$ with no action exchanging the pairs, and write

$$H = \hbar v k \sigma_z \otimes \tau_0 + m \sigma_x \otimes \tau_y. \quad (6)$$

The mass anticommutes with the kinetic term. Complex conjugation flips τ_y , while time reversal flips σ_x , so their product is even. This explicitly demonstrates why an even number of helical pairs need not be protected.

4 What is protected, and by what?

A nonmagnetic, time-reversal-symmetric one-body potential cannot elastically backscatter an isolated electron into its exact Kramers partner. This does not imply that every real edge is dissipationless: additional channels, inelastic processes, coupling between opposite edges, and interaction effects require separate analysis.

In three dimensions, band insulators in this symmetry class have one strong and three weak $\mathbb{Z}_2$ indices. A strong topological insulator has an odd number of surface Dirac cones in a symmetry-preserving band description. The weak indices additionally depend on lattice translation structure.

Takeaway. A statement about a protected boundary should name the protecting symmetry and the allowed perturbations. The discussion here assumes charge conservation and a band or weakly interacting description; strongly interacting surfaces can realize possibilities, including symmetry-preserving surface topological order, beyond a free-fermion mass test.

References

- [1] C. L. Kane and E. J. Mele, “ $\mathbb{Z}_2$ Topological Order and the Quantum Spin Hall Effect” (2005). [arXiv:cond-mat/0506581](#).
- [2] M. Z. Hasan and C. L. Kane, “Colloquium: Topological insulators,” *Rev. Mod. Phys.* **82**, 3045 (2010). [arXiv:1002.3895](#).